



# Jenkins

## Automated Deployment of Java Applications to Apache Tomcat :-

### Introduction :-

- Automated deployment is a key practice in CI/CD.
- It enables automatic build, test, and deployment of applications.
- Jenkins automates Java application deployment.
- Apache Tomcat is used to host Java applications.



### Jenkins :-

- Jenkins is an open-source automation tool written in Java.
- It supports building, testing, and deploying applications.
- Jenkins integrates with tools like Git, Maven, and Tomcat.
- It helps create automated CI/CD pipelines

### Apache Tomcat :-

- Apache Tomcat is an open-source web and application server.
- It implements Java Servlet, JSP, and WebSocket technologies.
- Tomcat is widely used for Java web applications.
- It deploys applications packaged as WAR files.

## Need for Automated Deployment :-

Manual deployment is time-consuming, error-prone, and inefficient. Automated deployment:

- Reduces human errors
- Improves deployment speed
- Ensures consistency across environments
- Enables faster feedback and releases

## Working Principle :-

1. Developers commit code to a version control system such as Git.
2. Jenkins detects code changes using webhooks or polling.
3. Jenkins pulls the latest code from the repository.
4. Maven compiles the source code and packages it into a WAR file.
5. Jenkins deploys the generated WAR file to Apache Tomcat.
6. Tomcat hosts the application and makes it available to users.

## Components Involved :-

1. **Version Control System (Git):** Stores application source code
2. **Jenkins:** Automates build and deployment process
3. **Maven:** Builds and packages Java applications
4. **Apache Tomcat:** Hosts and runs Java web applications

## Benefits :-

- Faster and reliable deployments
- Continuous integration and delivery
- Early detection of defects
- Improved developer productivity

## Limitations :-

- Initial setup complexity
- Requires proper plugin management
- Dependency on server availability
- Security configuration must be handled carefully

## **Real-World Applications :-**

- Enterprise Java web applications
- Microservices deployment
- DevOps pipelines
- Cloud-based CI/CD workflows

## Steps to Build & Deploy of Java Applications to Apache Tomcat :-

### Step — 1 : Install Git into your Jenkins server

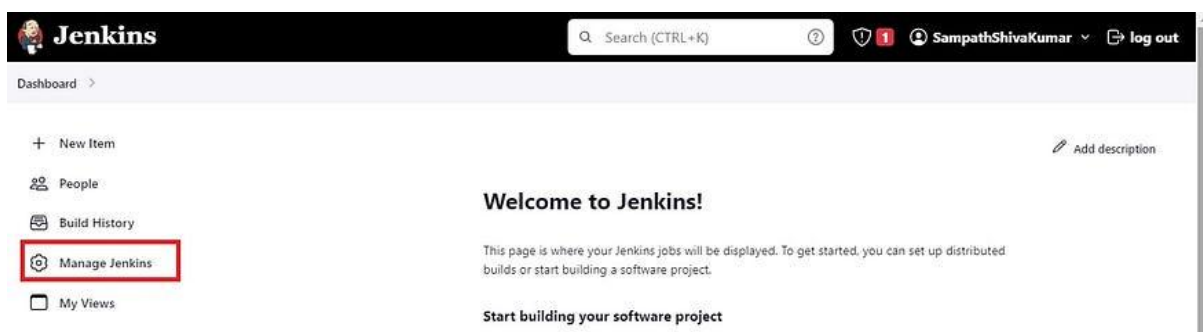
#### 1. install git and check the git version .

```
# Become a root  
sudo su -
```

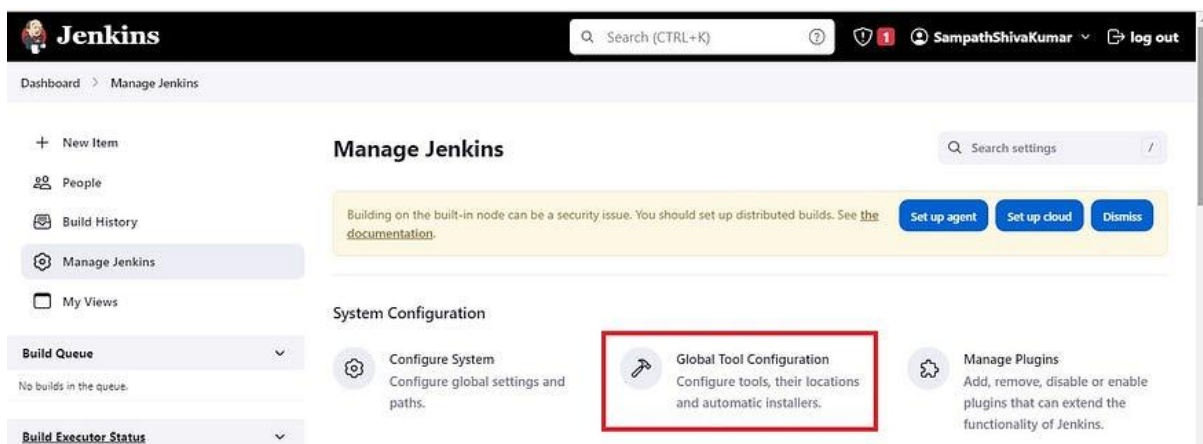
```
# Install git.  
yum install git -y
```

```
# Check Version of git.  
git --version
```

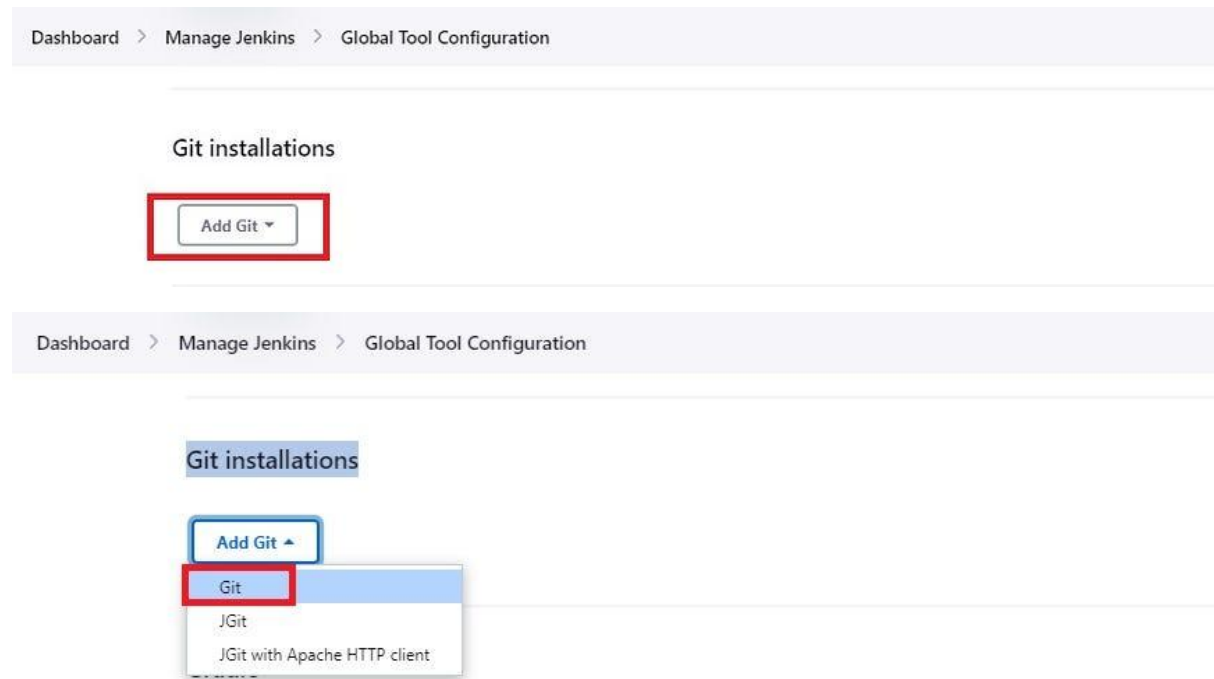
#### 2. Open Jenkins in web-browser, click on manage jenkins then select Global Tool Configuration.



#### 3. Then under Global tool configuration , add git in Git Installations



#### 4. In git installations , click Add git then select git.



#### 5. Enter the Git name and path , later click apply and save



**6. Now we have successfully integrated Git with Jenkins . Then click on Global Tool Configuration , under that go to Maven installtions . After giving the details click on apply and save .**

**Maven installations**

List of Maven installations on this system

Add Maven

1

≡ Maven

×

Name

maven

☒ Install automatically ?

≡ Install from Apache

×

Version

3.9.2

Add Installer

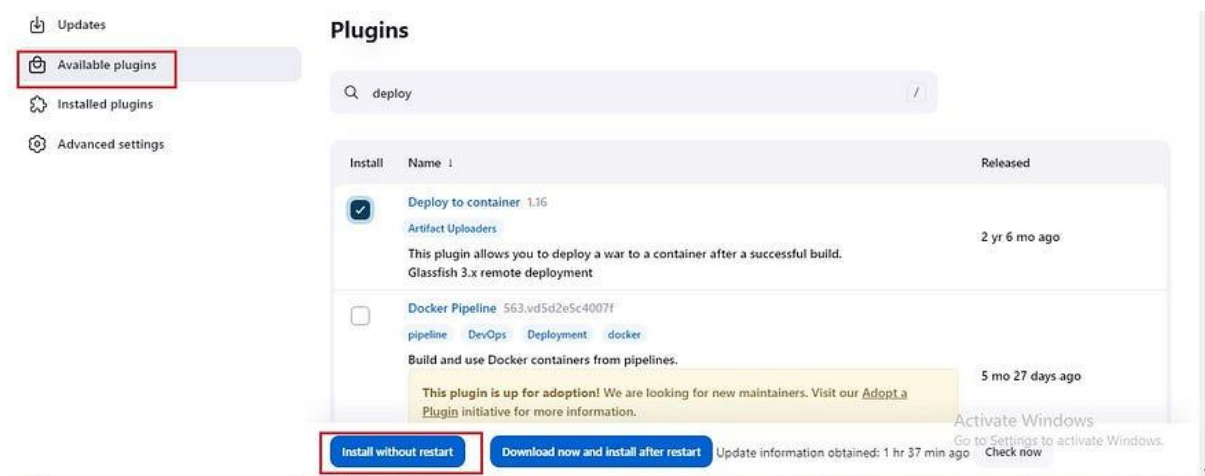
Save

Apply

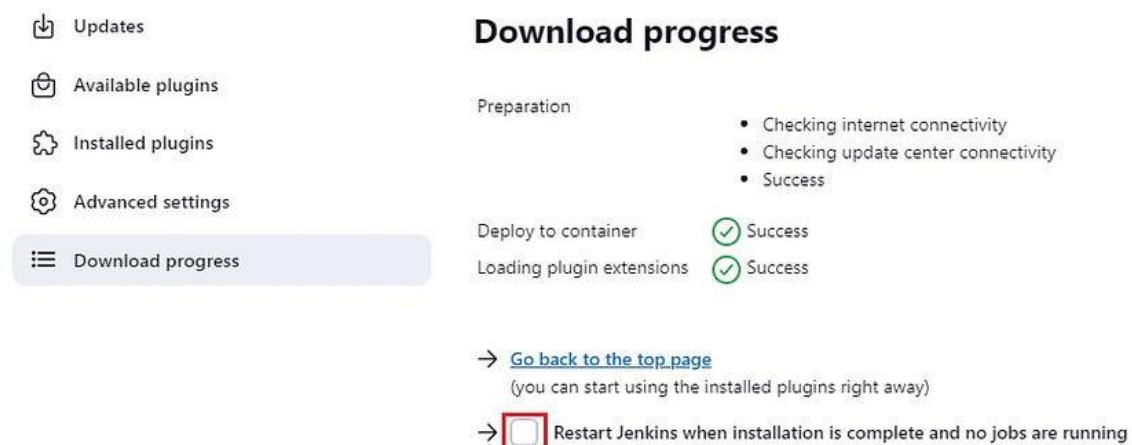
Now Maven is also successfully integrated with Jenkins .

## Steps — 2 : Install a Plugin called 'Deploy to container '

### 1. Go to manage jenkins > plugins > Available plugins > Deploy to container .



### 2. Click on the restart jenkins after installations , after the installations the jenkins will restart and you need to enter the username and password .



## step — 3 : Create a Global Credentials

1. Go to Credentials > Select system > Select Global Credentials then click on Add Credentials .

### Global credentials (unrestricted)

+ Add Credentials

Credentials that should be available irrespective of domain specification to requirements matching.

| ID                                                                                                   | Name                   | Kind        | Description                                                                                                |
|------------------------------------------------------------------------------------------------------|------------------------|-------------|------------------------------------------------------------------------------------------------------------|
|  slack-notification | slack-notification-key | Secret text | slack-notification-key  |

2. After clicking on add credentials , Enter the username and password

## Update credentials

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

tomcat

☐ Treat username as secret ?

Password ?

.....

ID ?

tomcat-deploy

Description ?

tomcat-deploy

Save

3. Once your done giviing the details , then you can see the credentails you have created .

Credentials

| T                                                                                 | P                                                                                 | Store ↓ | Domain   | ID                 | Name                              |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------|----------|--------------------|-----------------------------------|
|  |  | System  | (global) | tomcat-deploy      | tomcat/***** (tomcat-deploy)      |
|  |  | System  | (global) | tomcat-credentials | tomcat/***** (tomcat-credentials) |

## Step — 4: Create a Jenkins Job

1. Head to Dashboard > New Item , then provide a name for the jenkins job and select freestyle project as project type .

### New Item

Enter an item name

tomcat-jenkin

Select an item type



#### Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



#### Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



#### Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



#### Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



#### Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK

1. Enter the github repo and also change the branch name from master to main , because there is no master branch .

## Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/indrol12/DemoProjects.git

Credentials ?

- none -

Advanced ▼

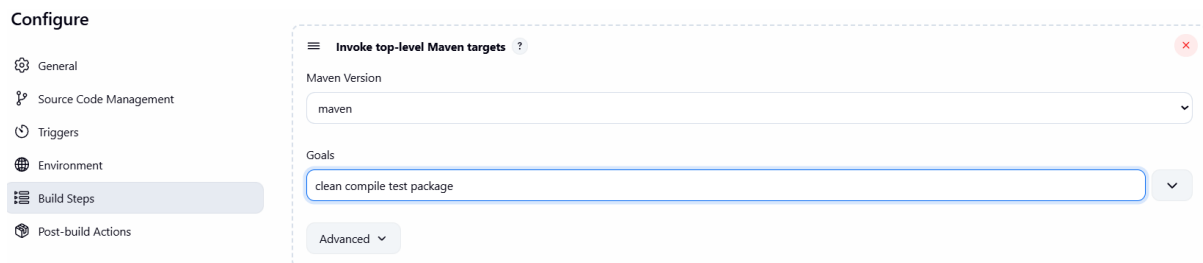
+ Add Repository

Branches to build ?

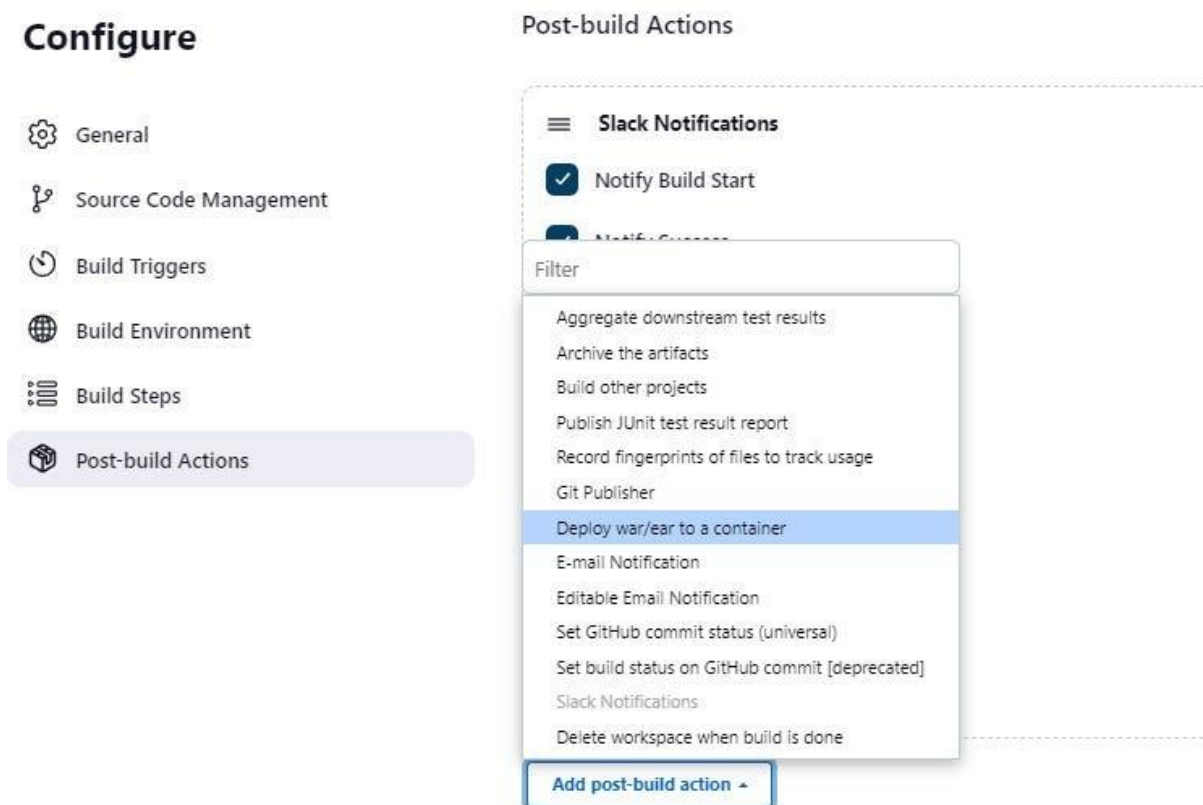
Branch Specifier (blank for 'any') ?

\*/master

3. Under Build steps > invoke top level maven targets , then provide maven command ' clean package' . And there is no need to mention mvn as we give in the terminal , it automatically takes as mvn and then runs the command .



4. In Post-build Actions click Add Post-build Actions and Select Deploy war/ear to container.



5. Enter the details as following then click on apply and save.

Deploy war/ear to a container

WAR/EAR files ?  

maven

Context path ?  

target\SpringFormApp.war

Containers

Tomcat 9.x Remote

Credentials  

admin/\*\*\*\*\* (this is tomcat creadinitalis)

+ Add

Tomcat URL ?  

http://localhost:1234

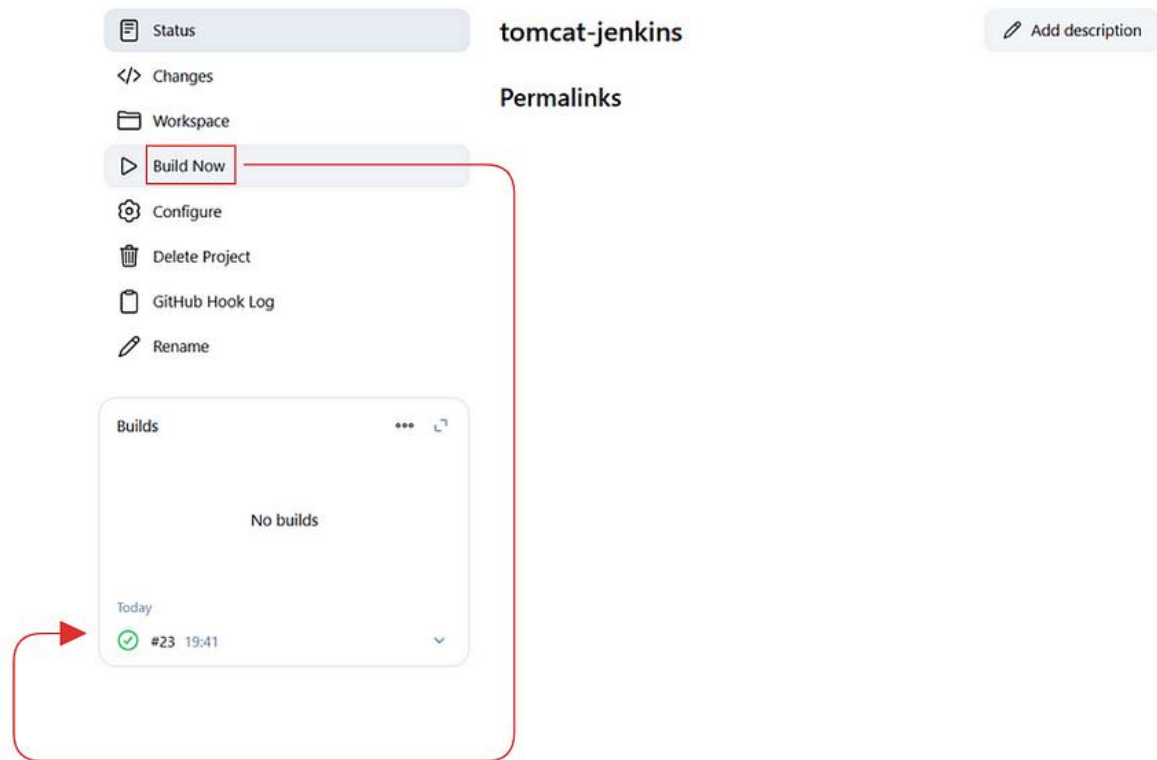
Advanced

+ Add Container

Save

Apply

Now Click on Build Now



The Build is Successfull .

## ✓ Console Output

Download

Copy

View as plain text

```
18:51:31 Started by timer
18:51:31 Running as SYSTEM
18:51:31 Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\build and deploy project
18:51:31 [WS-CLEANUP] Deleting project workspace...
18:51:31 [WS-CLEANUP] Deferred wipeout is used...
18:51:31 [WS-CLEANUP] Done
18:51:31 The recommended git tool is: NONE
18:51:31 using credential 87326e36-fbd1-478e-b31e-cd449829981e
18:51:31 Cloning the remote Git repository
18:51:31 Cloning repository https://github.com/indrol12/DemoProjects.git
18:51:31 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\build and deploy project #
timeout=10
18:51:32 Fetching upstream changes from https://github.com/indrol12/DemoProjects.git
18:51:32 > git.exe --version # timeout=10
```

## Output of the project in Apache Tomcat:-

### 1. Go to the localhost:1234 & click to Manager App.

[Home](#) [Documentation](#) [Configuration](#) [Examples](#) [Wiki](#) [Mailing Lists](#) [Find Help](#)

## Apache Tomcat/9.0.113



If you're seeing this, you've successfully installed Tomcat. Congratulations!

**Recommended Reading:**

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

[Server Status](#)

[Manager App](#)

[Host Manager](#)

### Developer Quick Start

[Tomcat Setup](#)  
[First Web Application](#)

[Realms & AAA](#)  
[JDBC DataSources](#)

[Examples](#)

[Servlet Specifications](#)  
[Tomcat Versions](#)

### Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager application is split between different users.  
[Read more...](#)

[Release Notes](#)

[Changelog](#)

[Migration Guide](#)

[Security Notices](#)

### Documentation

[Tomcat 9.0 Documentation](#)

[Tomcat 9.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 9.0 Bug Database](#)
- [Tomcat 9.0 JavaDocs](#)
- [Tomcat 9.0 Git Repository at GitHub](#)

### Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)  
Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)  
User support and discussion

[taglibs-user](#)  
User support and discussion for [Apache Taglibs](#)

[tomcat-dev](#)  
Development mailing list, including commit messages

### 2. Enter the username, password and submit then,



## Tomcat Web Application Manager

Message:  OK

### Manager

[List Applications](#)

[HTML Manager Help](#)

[Manager Help](#)

[Server Status](#)

### Applications

| Path            | Version        | Display Name                      | Running | Sessions | Commands                                                                                                                                                               |
|-----------------|----------------|-----------------------------------|---------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /               | None specified | Welcome to Tomcat                 | true    | 0        | <a href="#">Start</a> <a href="#">Stop</a> <a href="#">Reload</a> <a href="#">Undeploy</a><br><a href="#">Expire sessions</a> with idle ≥ <input type="text"/> minutes |
| /servletFormApp | None specified | Archetype Created Web Application | true    | 0        | <a href="#">Start</a> <a href="#">Stop</a> <a href="#">Reload</a> <a href="#">Undeploy</a><br><a href="#">Expire sessions</a> with idle ≥ <input type="text"/> minutes |
| /docs           | None specified | Tomcat Documentation              | true    | 0        | <a href="#">Start</a> <a href="#">Stop</a> <a href="#">Reload</a> <a href="#">Undeploy</a><br><a href="#">Expire sessions</a> with idle ≥ <input type="text"/> minutes |
| /manager        | None specified | Tomcat Manager Application        | true    | 1        | <a href="#">Start</a> <a href="#">Stop</a> <a href="#">Reload</a> <a href="#">Undeploy</a><br><a href="#">Expire sessions</a> with idle ≥ <input type="text"/> minutes |

### Deploy

Deploy directory or WAR file located on server

Context Path:

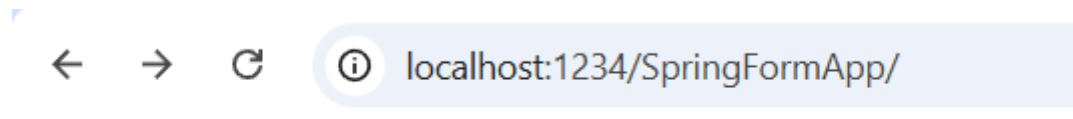
Version (for parallel deployment):

XML Configuration file path:

WAR or Directory path:

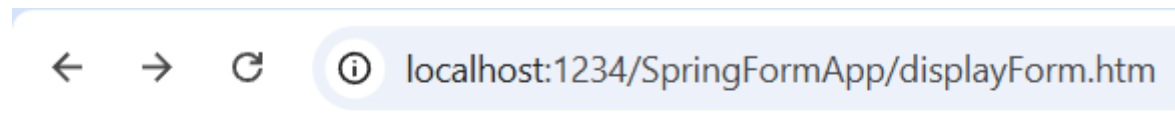
[Deploy](#)

### 3. Click on the /SpringFormApp



# Welcome to Fortune Cloud Technologies

[Registration Form](#)

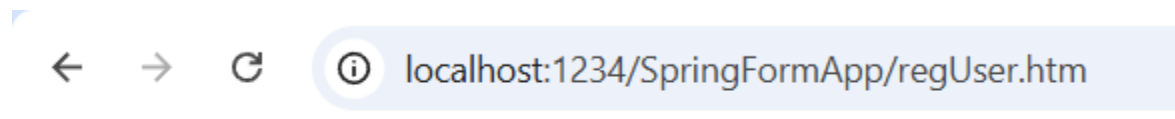


## Register here

Username:

Password:

Email:



## Registration successfull with below details

Username : abc  
Email : abc@gmail.com